Controller perfboard - build in stages

Star Destroyer 1/2700, Arduino Nano ESP32

Short stages you can do one session at a time. Each ends with tests: do not move on until every box is ticked. Board 1 carries the Nano and Driver C (the 7 engines); Board 2 carries Driver A and Driver B (hull, bridge and hangar bay). Build Board 1 completely first (Stages 1-6), then Board 2 (Stages 7-8).

Boards: two 5 x 7 cm prototype boards, 18 x 24 holes. **Hole names:** columns **A-X** across the top from left to right, rows **1-18** with **row 1 at the bottom**, seen from the component side. "K16" means column K, row 16. If your board prints its own grid differently, mark these names on it in pencil first.

Every channel plugs in with a JST XH 2-pin connector. On each header, **pin 1 = +5 V** (red) and **pin 2 = OUT** (black). On Board 1 the 7 engine sockets are **right-angle (side-entry)** XH sockets with their openings facing the bottom edge, plus an eighth right-angle socket for the accent lights in column T, opening facing the right-hand edge; on Board 2 the headers are upright, fitted the same way round with the lock ramp toward the bottom edge. Make every LED lead with red on plug pin 1, and check one lead with the meter: its red wire must reach the socket's +5 V pin. Then any LED or cluster plugs into any channel, so you can test each channel as soon as it is wired and swap LEDs on and off freely.

Test lead: one XH lead with a 330 ohm resistor and an LED on the end (long leg to the resistor on the red wire) - the same as an engine lead on page 8. Plug it into a header to prove that channel. Tools: multimeter with a continuity beep, and a bench 5 V supply (your BreadVolt).

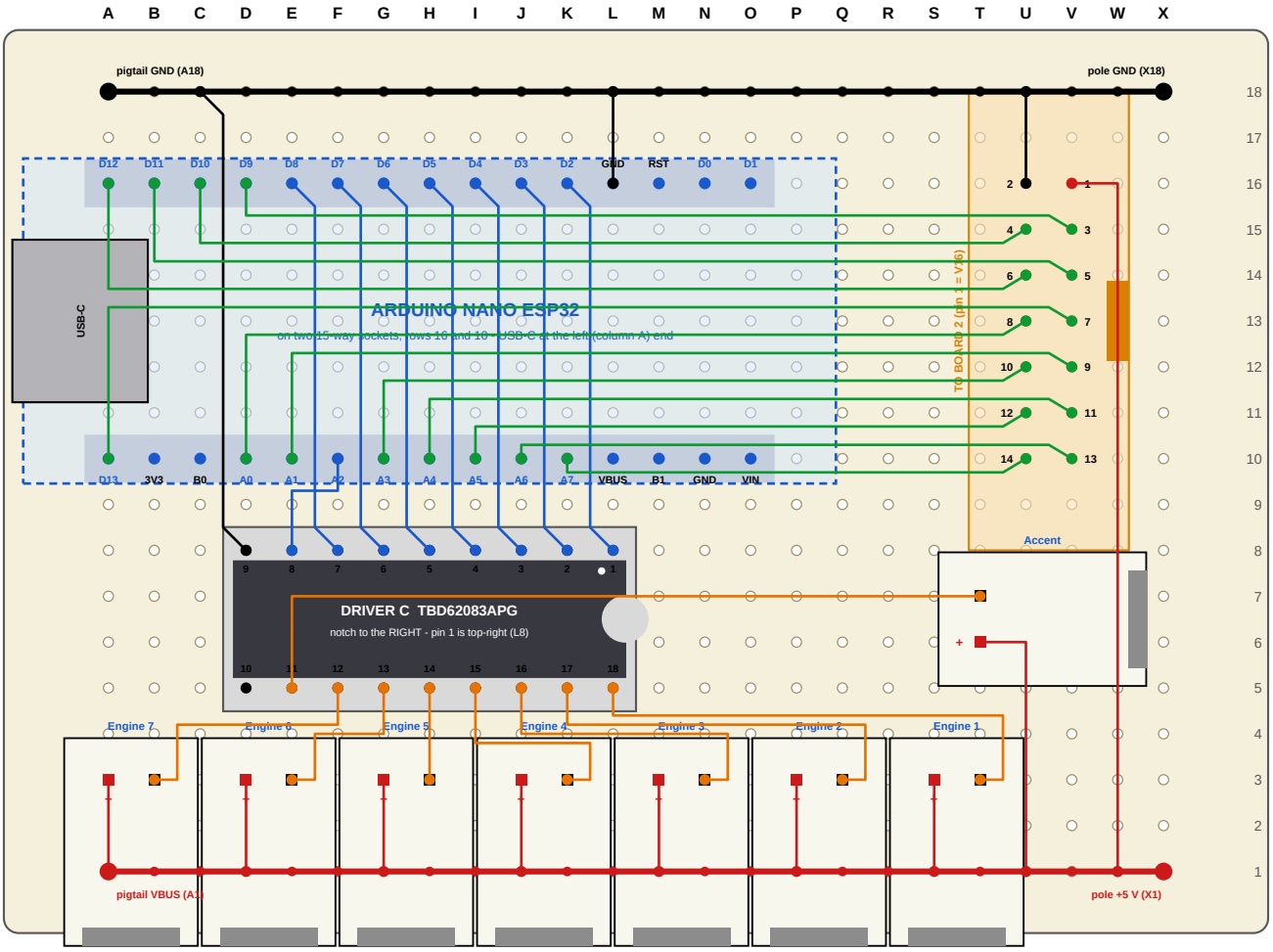
Lessons from the breadboard. Check ground with the beep test, never by voltage alone, because a floating pin can read 0 V. A poor ground shows up as the Nano resetting when many LEDs switch on at once.

Bus wires go in last. The two long bare power wires (GND along row 18, +5 V along row 1) are soldered only after every short wire is in place. Every other wire is thin **insulated** wire on the underside; insulated wires may cross each other, but never cross a bare bus wire.

Part (both boards)	Qty	Notes
5 x 7 cm prototype board, 18 x 24 holes, 2.54 mm pitch	2 + spare	Plain plated holes, no tracks
JST XH 2-pin right-angle socket (S2B-XH-A or compatible)	8 + spares	Board 1: 7 engines (opening to the bottom edge) and the accent socket (opening to the right)
JST XH 2-pin vertical header (B2B-XH-A or compatible)	12 + spares	Board 2
JST XH 2-pin lead, pre-crimped, red and black, 150-200 mm	21 + spares	One per cluster, one for the accent lights, plus the test lead
18-pin DIP socket / TBD62083APG	3 / 3	Plus spares
15-way female socket strip	2	For the Nano
14-way shrouded box header + 14-way IDC ribbon cable	2 + 1	Board-to-board harness
Thin insulated wire (28-30 AWG) and 0.6 mm tinned bare wire	-	Short wires and the two buses
Accent lights: stripboard, 6 x 1 k resistors, 6 pre-wired SMD LEDs (no built-in resistor)	2 boards / 6 / 6	Two junction boards, 3 x 11 holes (page 9)

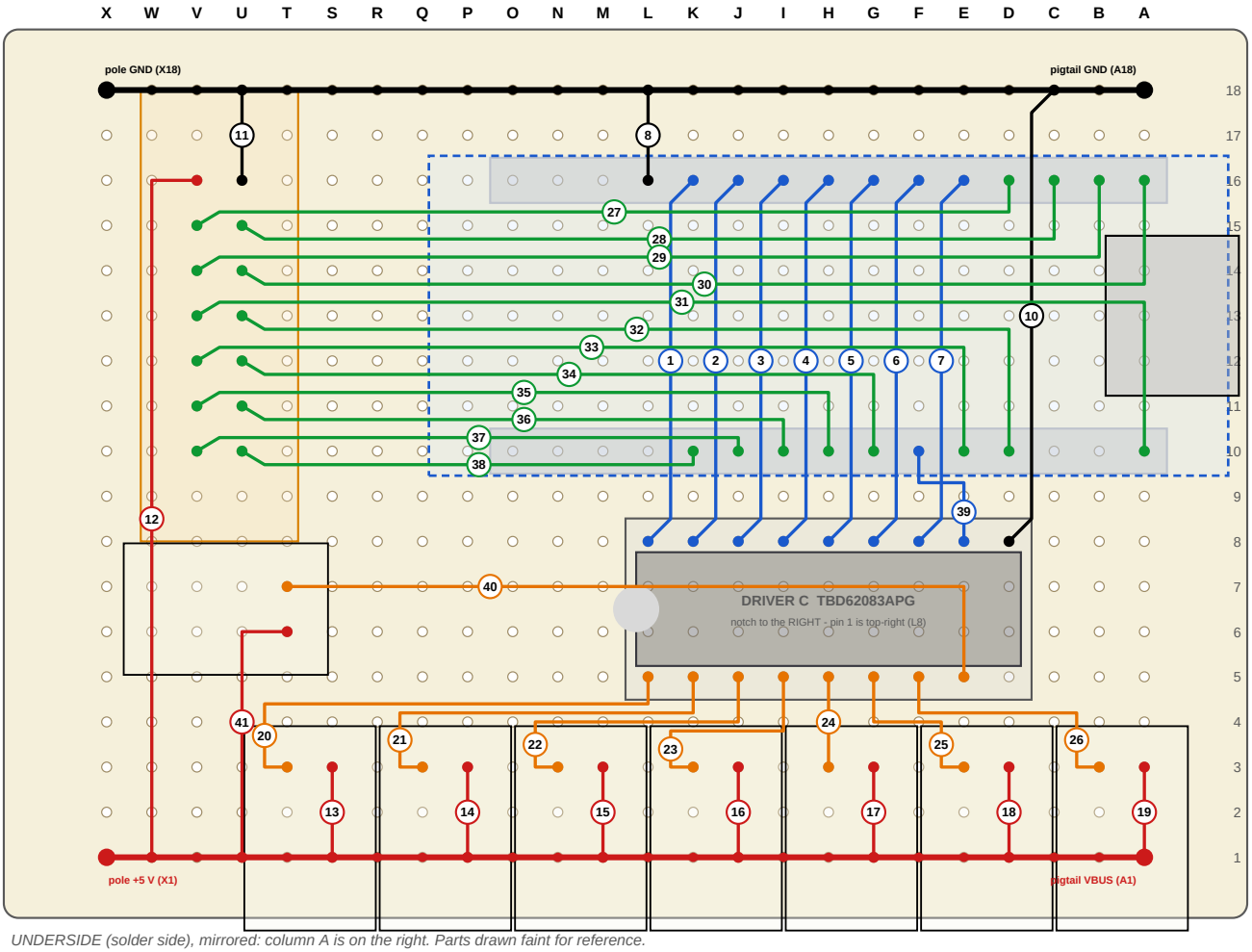
Board 1 - component side

Column A on the left. Nano on sockets in rows 16 and 10 (USB-C at the left edge). Driver C on its side in rows 8 and 5: **notch to the right, pin 1 at L8**. Engine sockets (right-angle, opening toward the bottom edge) along row 3 - Engine 7 on the left (A-B) to Engine 1 on the right (S-T); +5 V pins in A, D, G, J, M, P, S. Box header at U-V, pin 1 at V16. **Accent socket** (right-angle XH) in column T, between Driver C and the box header, its opening facing the right-hand edge: pin 1 (+5 V) at T6, pin 2 (OUT) at T7. This keeps the plug clear of the thick +5 V wire from the box header down column W.



Board 1 - underside

Turned over like a page, so column A is on the right. Each wire has its number from the table.



UNDERSIDE (solder side), mirrored: column A is on the right. Parts drawn faint for reference.

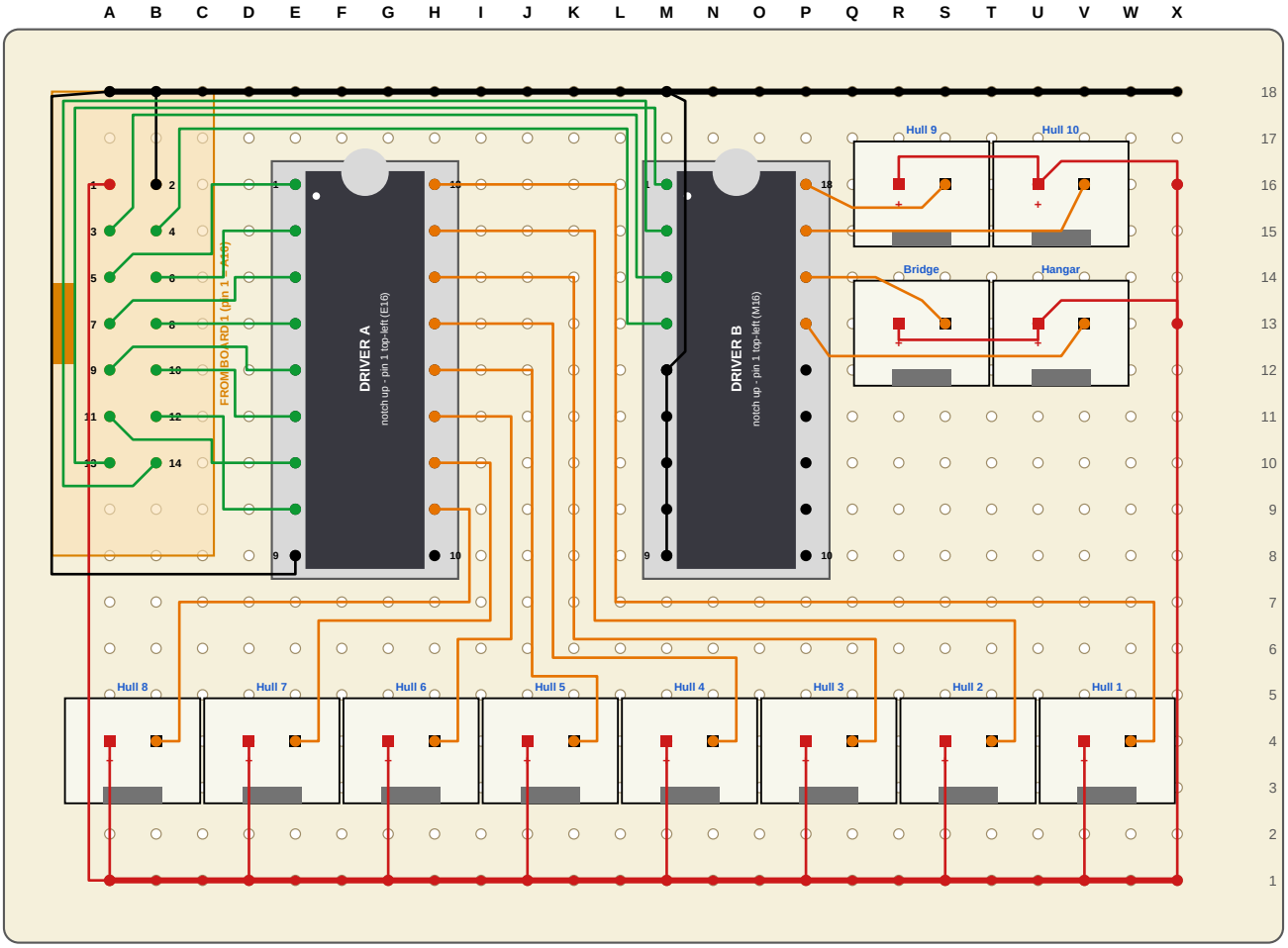
Board 1 - wires to fit (tick each one)

Buses (bare wire, laid last): **GND along row 18** and **+5 V along row 1**, A to X. Pole wires: X1 (+5 V) and X18 (GND). USB-C power pigtail: A1 (VBUS) and A18 (GND). Driver C pin 10 (D5, COMMON) stays unconnected; Nano B0, B1, 3V3, VBUS and VIN are not used. There is **no wire 9**: Driver C input 8 is now the accent channel (wires 39-41).

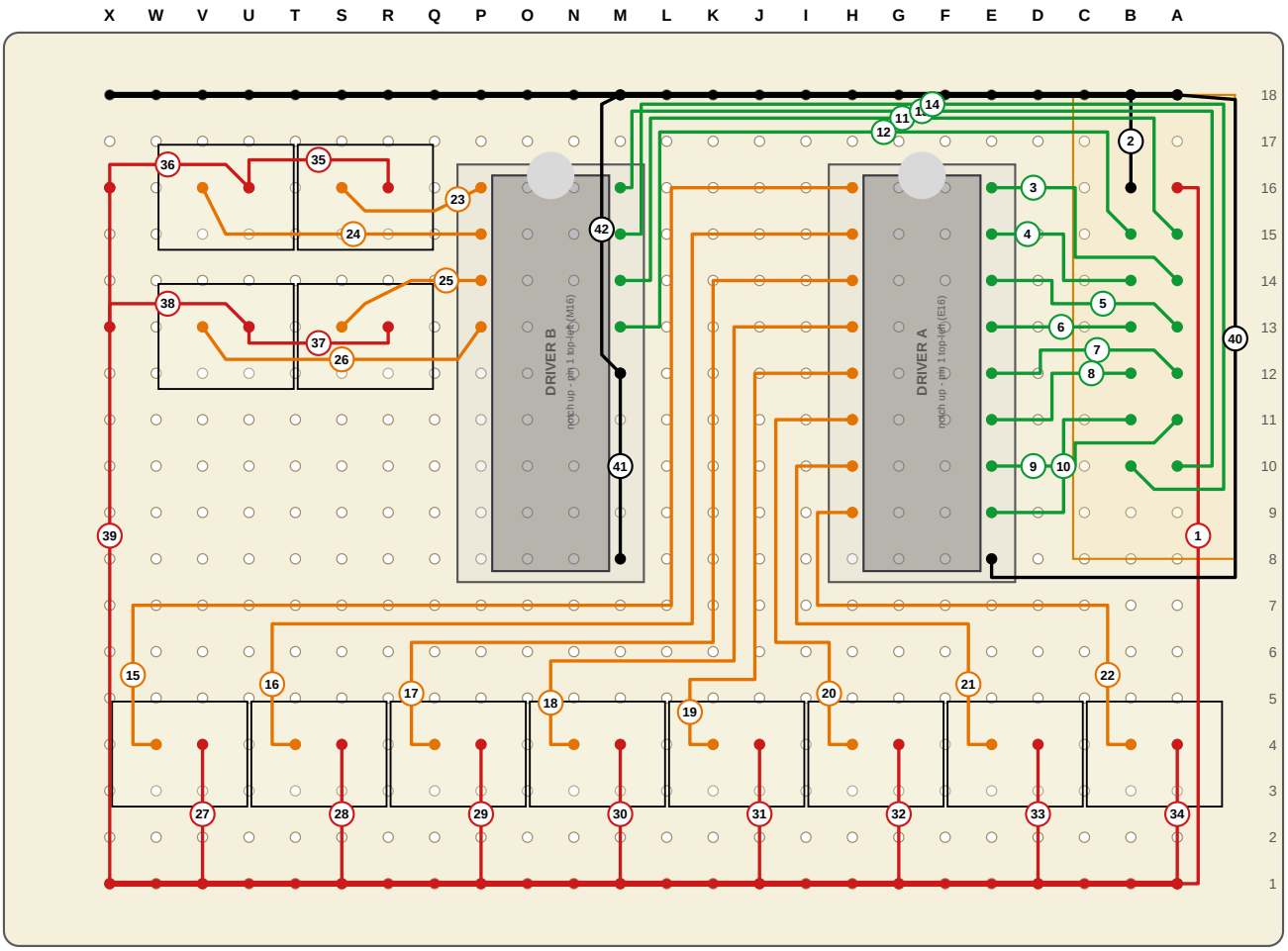
Tick	#	Colour	What it joins	From	To
[ ]	1	Blue	Nano D2 to Driver C pin 1 (engine 1)	K16	L8
[ ]	2	Blue	Nano D3 to Driver C pin 2 (engine 2)	J16	K8
[ ]	3	Blue	Nano D4 to Driver C pin 3 (engine 3)	I16	J8
[ ]	4	Blue	Nano D5 to Driver C pin 4 (engine 4)	H16	I8
[ ]	5	Blue	Nano D6 to Driver C pin 5 (engine 5)	G16	H8
[ ]	6	Blue	Nano D7 to Driver C pin 6 (engine 6)	F16	G8
[ ]	7	Blue	Nano D8 to Driver C pin 7 (engine 7)	E16	F8
[ ]	8	Black	Nano GND to the GND bus	L16	L18
[ ]	10	Black	Driver C pin 9 (GND) to the GND bus	D8	C18
[ ]	11	Black	Box header pin 2 (GND) to the GND bus	U16	U18
[ ]	12	Red	Box header pin 1 (+5 V) to the +5 V bus	V16	W1
[ ]	13	Red	Engine 1 socket pin 1 (+5 V) to the +5 V bus	S3	S1
[ ]	14	Red	Engine 2 socket pin 1 (+5 V) to the +5 V bus	P3	P1
[ ]	15	Red	Engine 3 socket pin 1 (+5 V) to the +5 V bus	M3	M1
[ ]	16	Red	Engine 4 socket pin 1 (+5 V) to the +5 V bus	J3	J1
[ ]	17	Red	Engine 5 socket pin 1 (+5 V) to the +5 V bus	G3	G1
[ ]	18	Red	Engine 6 socket pin 1 (+5 V) to the +5 V bus	D3	D1
[ ]	19	Red	Engine 7 socket pin 1 (+5 V) to the +5 V bus	A3	A1
[ ]	20	Orange	Driver C pin 18 to Engine 1 socket pin 2 (OUT)	L5	T3
[ ]	21	Orange	Driver C pin 17 to Engine 2 socket pin 2 (OUT)	K5	Q3
[ ]	22	Orange	Driver C pin 16 to Engine 3 socket pin 2 (OUT)	J5	N3
[ ]	23	Orange	Driver C pin 15 to Engine 4 socket pin 2 (OUT)	I5	K3
[ ]	24	Orange	Driver C pin 14 to Engine 5 socket pin 2 (OUT)	H5	H3
[ ]	25	Orange	Driver C pin 13 to Engine 6 socket pin 2 (OUT)	G5	E3
[ ]	26	Orange	Driver C pin 12 to Engine 7 socket pin 2 (OUT)	F5	B3
[ ]	27	Green	Nano D9 to box header pin 3	D16	V15
[ ]	28	Green	Nano D10 to box header pin 4	C16	U15
[ ]	29	Green	Nano D11 to box header pin 5	B16	V14
[ ]	30	Green	Nano D12 to box header pin 6	A16	U14
[ ]	31	Green	Nano D13 to box header pin 7	A10	V13
[ ]	32	Green	Nano A0 to box header pin 8	D10	U13
[ ]	33	Green	Nano A1 to box header pin 9	E10	V12
[ ]	34	Green	Nano A3 to box header pin 10	G10	U12
[ ]	35	Green	Nano A4 to box header pin 11	H10	V11
[ ]	36	Green	Nano A5 to box header pin 12	I10	U11
[ ]	37	Green	Nano A6 to box header pin 13	J10	V10
[ ]	38	Green	Nano A7 to box header pin 14	K10	U10
[ ]	39	Blue	Nano A2 to Driver C pin 8 (accent lights)	F10	E8
[ ]	40	Orange	Driver C pin 11 to Accent socket pin 2 (OUT)	E5	T7
[ ]	41	Red	Accent socket pin 1 (+5 V) to the +5 V bus at U1	T6	U1

Board 2 - component side

Driver A in columns E-H and Driver B in columns M-P, rows 16-8, both **notch at the top**. Box header at A-B, pin 1 at A16. Hull 1-8 headers along row 4 (Hull 8 on the left to Hull 1 on the right). Hull 9 and Hull 10 at R-S and U-V on row 16; Bridge and Hangar at R-S and U-V on row 13.



Board 2 - underside



UNDERSIDE (solder side), mirrored: column A is on the right. Parts drawn faint for reference.

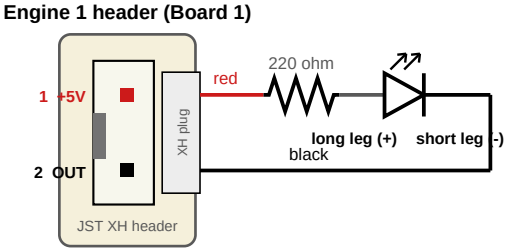
Board 2 - wires to fit (tick each one)

Buses (bare wire, laid last): **GND along row 18** and **+5 V along row 1**, A to X. Driver A pin 10 (H8) and Driver B pins 10-14 (P8-P12) stay unconnected.

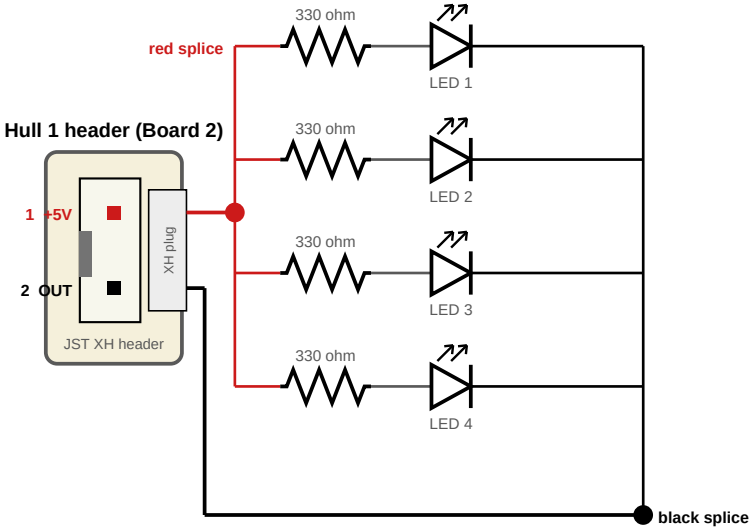
Tick	#	Colour	What it joins	From	To
[ ]	1	Red	Box header pin 1 (+5 V) to the +5 V bus, down the left edge	A16	A1
[ ]	2	Black	Box header pin 2 (GND) to the GND bus	B16	B18
[ ]	3	Green	Box header pin 5 (D11) to Driver A pin 1 (Hull 1)	A14	E16
[ ]	4	Green	Box header pin 6 (D12) to Driver A pin 2 (Hull 2)	B14	E15
[ ]	5	Green	Box header pin 7 (D13) to Driver A pin 3 (Hull 3)	A13	E14
[ ]	6	Green	Box header pin 8 (A0) to Driver A pin 4 (Hull 4)	B13	E13
[ ]	7	Green	Box header pin 9 (A1) to Driver A pin 5 (Hull 5)	A12	E12
[ ]	8	Green	Box header pin 10 (A3) to Driver A pin 6 (Hull 6)	B12	E11
[ ]	9	Green	Box header pin 11 (A4) to Driver A pin 7 (Hull 7)	A11	E10
[ ]	10	Green	Box header pin 12 (A5) to Driver A pin 8 (Hull 8)	B11	E9
[ ]	11	Green	Box header pin 3 (D9) to Driver B pin 3 (Bridge)	A15	M14
[ ]	12	Green	Box header pin 4 (D10) to Driver B pin 4 (Hangar)	B15	M13
[ ]	13	Green	Box header pin 13 (A6) to Driver B pin 1 (Hull 9)	A10	M16
[ ]	14	Green	Box header pin 14 (A7) to Driver B pin 2 (Hull 10)	B10	M15
[ ]	15	Orange	Driver A pin 18 to Hull 1 header pin 2 (OUT)	H16	W4
[ ]	16	Orange	Driver A pin 17 to Hull 2 header pin 2 (OUT)	H15	T4
[ ]	17	Orange	Driver A pin 16 to Hull 3 header pin 2 (OUT)	H14	Q4
[ ]	18	Orange	Driver A pin 15 to Hull 4 header pin 2 (OUT)	H13	N4
[ ]	19	Orange	Driver A pin 14 to Hull 5 header pin 2 (OUT)	H12	K4
[ ]	20	Orange	Driver A pin 13 to Hull 6 header pin 2 (OUT)	H11	H4
[ ]	21	Orange	Driver A pin 12 to Hull 7 header pin 2 (OUT)	H10	E4
[ ]	22	Orange	Driver A pin 11 to Hull 8 header pin 2 (OUT)	H9	B4
[ ]	23	Orange	Driver B pin 18 to Hull 9 header pin 2 (OUT)	P16	S16
[ ]	24	Orange	Driver B pin 17 to Hull 10 header pin 2 (OUT)	P15	V16
[ ]	25	Orange	Driver B pin 16 to Bridge header pin 2 (OUT)	P14	S13
[ ]	26	Orange	Driver B pin 15 to Hangar header pin 2 (OUT)	P13	V13
[ ]	27	Red	Hull 1 header pin 1 (+5 V) to the +5 V bus	V4	V1
[ ]	28	Red	Hull 2 header pin 1 (+5 V) to the +5 V bus	S4	S1
[ ]	29	Red	Hull 3 header pin 1 (+5 V) to the +5 V bus	P4	P1
[ ]	30	Red	Hull 4 header pin 1 (+5 V) to the +5 V bus	M4	M1
[ ]	31	Red	Hull 5 header pin 1 (+5 V) to the +5 V bus	J4	J1
[ ]	32	Red	Hull 6 header pin 1 (+5 V) to the +5 V bus	G4	G1
[ ]	33	Red	Hull 7 header pin 1 (+5 V) to the +5 V bus	D4	D1
[ ]	34	Red	Hull 8 header pin 1 (+5 V) to the +5 V bus	A4	A1
[ ]	35	Red	Hull 9 pin 1 to Hull 10 pin 1 (+5 V link)	R16	U16
[ ]	36	Red	Hull 10 pin 1 to the right edge	U16	X16
[ ]	37	Red	Bridge pin 1 to Hangar pin 1 (+5 V link)	R13	U13
[ ]	38	Red	Hangar pin 1 to the right edge	U13	X13
[ ]	39	Red	Right-edge +5 V wire down to the +5 V bus	X16	X1
[ ]	40	Black	Driver A pin 9 (GND) to the GND bus, up the left edge	E8	A18
[ ]	41	Black	Driver B unused inputs 5-8 joined to pin 9	M12	M8
[ ]	42	Black	Driver B pins 5-9 (GND) to the GND bus	M12	M18

How a channel is wired

Every channel is a 2-pin JST XH header: **pin 1 = +5 V** (from the +5 V bus) and **pin 2 = OUT** (from a driver output). The LED lead ends in an XH plug: red wire to pin 1, black to pin 2. When the sketch turns the channel on, the driver connects OUT to ground and current flows: pin 1, resistor, LED long leg, LED short leg, pin 2, driver, GND. The resistor sets the current, so every LED has its own.



One LED and its 220 ohm resistor, no splice: red to plug pin 1, black to plug pin 2. Engines 2-7 are the same; the test lead is this with 330 ohm.

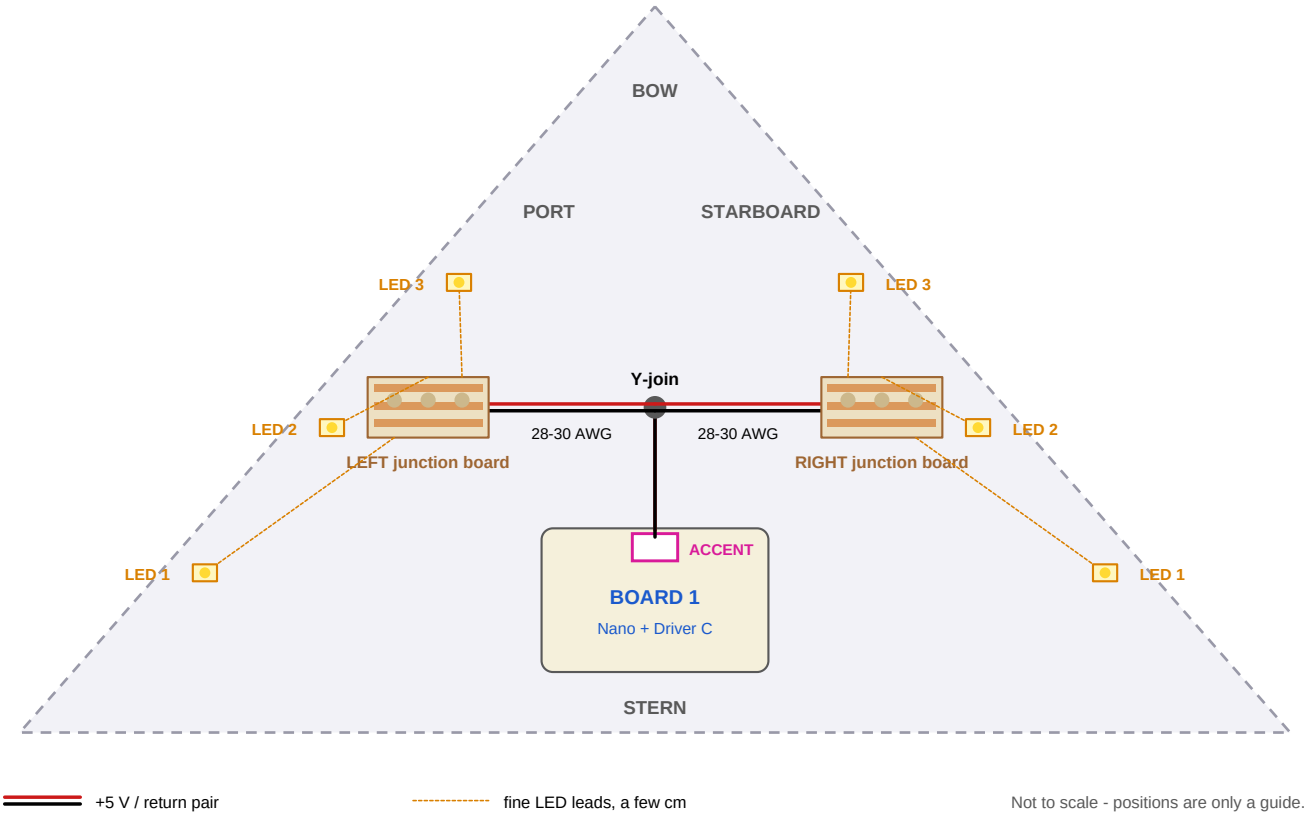


Each LED gets its own resistor on its long (+) leg. All 4 red leads join the red wire to plug pin 1 (+5V); all 4 black leads join the black wire to plug pin 2 (OUT). Solder every splice and cover it in heat-shrink. The bridge is the same with 2 LEDs; the hangar bay has 4.

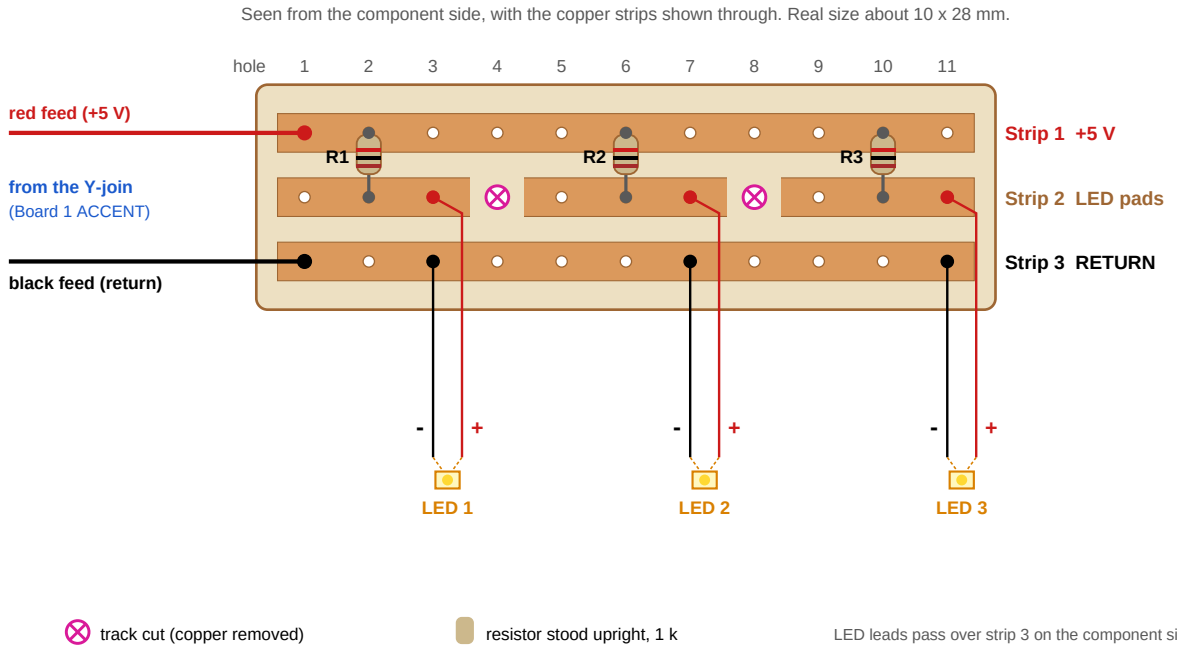
Engines 1-7 on Board 1 are the left drawing. The hull zones, bridge and hangar bay on Board 2 are the right-hand drawing. If an LED is in backwards, its channel stays dark but nothing is damaged: turn it round.

Channel 20 - the accent lights

Six SMD LEDs, three along each side of the hull, on one channel: Nano **A2** drives Driver C input 8, and its output (pin 11) goes to the **Accent** socket on the right-hand edge of Board 1. One lead from that header splits at a Y-join into two red/black pairs (28-30 AWG), one to a small **junction stripboard** on each side. Each junction board carries three 1 k resistors, and the LEDs' fine pre-wired leads run only a few centimetres from the board. All six fade up together as a steady glow, 14 s into the show.



Each junction board is 3 strips by 11 holes. Strip 1 is +5 V, strip 3 is the return, and strip 2 is cut at holes 4 and 8 to make three separate LED pads:



Before fitting each junction board: beep test that strip 1 does not beep to strip 3 and the three pads are separate; then power it from the bench supply (**+5v** to the red feed, **negative** to the black feed) and check all three LEDs light. Cover the copper side with Kapton tape, glue it in, and tack the fine leads down with a slack loop at each LED. Start with 1 k resistors; use 470 R if the LEDs look too dim.

Stage 1 - Board 1: mark out and dry-fit

Goal: Know exactly where everything goes, before any solder.

Do

- [ ] Mark the hole names and the positions from the Board 1 drawing in pencil.
- [ ] Without soldering, place the two 15-way Nano sockets (rows 16 and 10, columns A-O), the DIP socket (rows 8 and 5, columns D-L, notch to the right), the box header (columns U-V, rows 16-10, pin 1 at V16) and the 7 right-angle XH sockets along row 3, openings facing the bottom edge, and the right-angle Accent socket at T6-T7 with its opening facing the right-hand edge.
- [ ] Fit the Nano: its USB-C end is at the left edge, over column A.

Tests - all must pass before the next stage

- [ ] Every part sits flat on the marked holes; no XH header touches its neighbour.
- [ ] The Nano clears the box header and the DIP socket, and a USB-C plug fits.

Stage 2 - Board 1: sockets and headers

Goal: The Nano sockets, the Driver C socket, the box header, the 7 engine sockets and the Accent socket are soldered flat, with no wires yet.

Do

- [ ] Solder each part: tack one pin, check it is flat and square, then do the rest. Leave the chip and the Nano out.
- [ ] XH sockets: pin 1 (+5 V) in columns A, D, G, J, M, P, S and pin 2 (OUT) in B, E, H, K, N, Q, T, all on row 3.
- [ ] Accent socket (right-angle): pin 1 (+5 V) at T6, pin 2 (OUT) at T7, opening facing the right-hand edge. Check a plug goes in freely before soldering.

Tests - all must pass before the next stage

- [ ] Everything sits flat on the board, with no gap or tilt.
- [ ] Beep test: no two neighbouring pins beep to each other (no solder bridges).

Stage 3 - Board 1: short power and ground wires

Goal: Every ground and +5 V connection is wired to its bus row; the two bus wires are NOT laid yet.

Do

- [ ] Ground (wires 8, 10, 11 - there is no wire 9): Nano GND L16 to L18; Driver C pin 9 (D8) up between columns C and D to C18; box header pin 2 U16 to U18. Driver C pin 8 (E8) is **not** grounded: it is the accent input.
- [ ] +5 V (wires 12-19 and 41): box header pin 1 V16 across to W16 and down column W to W1; each engine socket pin 1 straight down to row 1 (A3 to A1, D3 to D1 ... S3 to S1); Accent socket pin 1 (T6) one hole right to U6, then down column U to U1.
- [ ] End each wire at its bus-row hole on the underside, with a short bare tip ready for the bus wire.

Tests - all must pass before the next stage

- [ ] Beep test: each wire beeps from end to end.
- [ ] Beep test between any +5 V wire and any GND wire: **must NOT beep**.

Stage 4 - Board 1: bus wires last, then power up

Goal: The two bus wires are in, the board powers up, and the Nano runs.

Do

- [ ] Lay a straight bare tinned wire (about 0.6 mm) along row 18 (GND) and row 1 (+5 V), A to X, and solder it at every hole.
- [ ] Solder the bench power leads (later the pole wires, through the fuse) to X1 (+5 V) and X18 (GND). Use plenty of solder; the ground joint must be strong.
- [ ] Make the USB-C power pigtail from A1 (VBUS) and A18 (GND). Never plug it in while the Nano is connected to your computer.

Tests - all must pass before the next stage

- [ ] Beep test: along the whole +5 V bus and the whole GND bus; the two buses must **NOT** beep to each other.
- [ ] Bench supply 5 V: 5 V between the buses at both ends, and the supply current is close to zero.
- [ ] Every engine socket and the Accent socket: 5 V between pin 1 and the GND bus.
- [ ] With only the computer cable, upload Blink: the small orange LED flashes. Then, powered from the pigtail alone, the Nano starts up (RGB LED lit). Unplug it again.

Stage 5 - Board 1: signal wires

Goal: Driver C and the box header are wired to the Nano, and the chip is fitted.

Do

- [ ] **Green wires first.** Wires 27-38: the 12 green wires from the Nano to the box header, one at a time from the table. Several of them solder to the Nano's bottom row of pins (row 10), so do them all before the blue wires, which pass between those pins.
- [ ] **Then the blue wires.** Wires 1-7: Nano D2 (K16) to Driver C pin 1 (L8), D3 (J16) to pin 2 (K8), and so on to D8 (E16) to pin 7 (F8). Each runs between two columns, so it never passes over another pin. They are only soldered at row 16 and row 8, never at row 10. If you need to re-solder a row 10 pin later, lay a strip of Kapton tape over the blue wires first.
- [ ] Wire 39: Nano A2 (F10) one hole down, across to column E and into Driver C pin 8 (E8). It crosses the D8 wire; both are insulated, so that is fine.
- [ ] Fit the chip: **notch to the right**, pin 1 at L8.

Tests - all must pass before the next stage

- [ ] Beep test: each wire end to end, and no wire beeps to its neighbour.
- [ ] No chip leg is bent under the body, and the notch is to the right.

Stage 6 - Board 1: outputs and the engines

Goal: All 7 engine channels and the accent channel switch a real load. This proves the whole Board 1 path.

Do

- [ ] Wires 20-26: Driver C pin 18 (L5) to Engine 1 pin 2 (T3), pin 17 (K5) to Engine 2 (Q3), and so on to pin 12 (F5) to Engine 7 (B3), as drawn (no crossings). Pin 14 (H5) runs straight down to Engine 5 (H3).
- [ ] Wire 40: Driver C pin 11 (E5) up to row 7 (between the chip's pin rows), then right along row 7, under the chip, to the Accent socket pin 2 (T7).

Tests - all must pass before the next stage

- [ ] Beep test: each output to its header pin 2; no output beeps to another output or to GND.
- [ ] **Take the Nano out of its sockets first** (its pins are 3.3 V only and +5 V would damage them). Plug the test lead into Engine 1. Hold a wire from +5 V to Driver C pin 1 for a moment: the LED lights. Repeat for Engines 2-7 with pins 2-7.
- [ ] Still with the Nano out: plug the test lead into the Accent socket and hold +5 V on Driver C pin 8 (E8) for a moment: the LED lights. Refit the Nano.

- [ ] Plug in all 7 engine LEDs and upload **engine_test_nano.ino**: one at a time, all together, fade each, production look.
- [ ] **All together test:** in step 2 all 7 light with no Nano reset. A reset means a poor ground joint.

Stage 7 - Board 2: sockets, headers and power

Goal: Board 2 has its parts, its power and ground wiring and its buses, and gets power over the ribbon cable.

Do

- [ ] Solder the two DIP sockets upright, notch at the top (Driver A in columns E-H, Driver B in columns M-P, rows 16-8), the box header (pin 1 at A16) and the 12 XH headers (lock ramp toward the bottom edge): Hull 1-8 along row 4, Hull 9 and Hull 10 on row 16, Bridge and Hangar on row 13.
- [ ] Power and ground wires (1, 2, 27-39, 40-42): header pin 1 A16 down the left edge to A1; header pin 2 B16 to B18; each Hull 1-8 header pin 1 straight down to row 1; the four right-hand headers' pin 1s joined and run down column X to X1; Driver A pin 9 (E8) round the left edge to A18; Driver B pins 5-9 (M12 to M8) joined and run up to M18.
- [ ] Buses last: bare wire along row 18 (GND) and row 1 (+5 V), A to X.

Tests - all must pass before the next stage

- [ ] Beep tests: no bridges; each wire end to end; the +5 V bus does **not** beep to the GND bus.
- [ ] Plug the ribbon cable between the two boards (red stripe to pin 1 at both ends) and power Board 1: Board 2 reads 5 V between its buses and 5 V on every XH header's pin 1.

Stage 8 - Board 2: signals, outputs and all 20 channels

Goal: Every channel on both boards works from the Nano.

Do

- [ ] Wires 3-14: the 12 green wires from the box header to the driver inputs, from the table.
- [ ] Wires 15-26: Driver A pins 18-11 (H16 to H9) to the Hull 1-8 headers' pin 2, and Driver B pins 18-15 (P16 to P13) to the Hull 9, Hull 10, Bridge and Hangar headers' pin 2.
- [ ] Fit both chips, notch at the top.

Tests - all must pass before the next stage

- [ ] Beep test through the ribbon cable: each Nano pin on Board 1 to its driver input on Board 2 (for example Nano D11 to Driver A pin 1 at E16).
- [ ] Upload **pin_test_nano.ino** and plug the test lead into each header in turn: every channel lights in its turn, ending with the Accent socket on Board 1.
- [ ] Nothing lights that should be off, and the Nano does not reset at any point.

Stage 9 - Full system and soak

Goal: The finished electronics are ready to go into the hull.

Do

- [ ] Plug in all 19 LED clusters (Stage 3 of the build guide) and the accent lead with both junction boards, then upload the main sketch.
- [ ] Run the show from the Bluefruit Connect app, ON then OFF.

Tests - all must pass before the next stage

- [ ] All 20 channels behave correctly in the show; engines breathe and shimmer, and the accent lights fade up to a steady glow.
- [ ] Run for at least 30 minutes with no resets and no hot parts (feel the chips and resistors).
- [ ] Check the supply current stays within your fuse rating. Finish all of this before closing the hull.